Lab Tasks

Task # 01:

Create a Java program for a banking system using single inheritance. The base class should be called "BankAccount" and should have:

- String accountNumber
- String accountHolderName
- private double balance
- void deposit(double amount) - adds the given amount to the account balance
- void withdraw(double amount) - subtracts the given amount from the account balance

The subclass should be called "SavingsAccount" and should inherit from the BankAccount class. The SavingsAccount class should have an additional property called "interestRate" (double), which represents the annual interest rate for the account.

- The SavingsAccount class should also have a method called "calculateInterest()", which calculates the interest earned on the account balance using the interestRate and adds it to the account balance.

Create an instance of the SavingsAccount class, set its properties (including those inherited from BankAccount), and then perform a series of transactions on the account (e.g. deposit some money, withdraw some money, calculate interest, etc.). Finally, the program should print out the details of the account, including the account number, account holder name, balance, and interest rate.

Task # 02:

Create a Java program for a university course registration system using with multi-level inheritance. The base class should be called "Course" and should have the following: properties and methods:

- protected String courseCode
- protected String courseName
- protected int courseCredits
- void setCourseCode(String courseCode) - sets the course code
- void setCourseName(String courseName) - sets the course name
- void setCourseCredits(int courseCredits) - sets the course credits

The first subclass should be called "UndergraduateCourse" and should inherit from the Course class. The UndergraduateCourse class should have the following additional properties and methods:

- protected String departmentName

- protected int level (e.g. 1 for freshman, 2 for sophomore, etc.)
- void setDepartmentName(String departmentName) - sets the department name
- void setLevel(int level) - sets the level of the undergraduate course.

The second subclass should be called "Registration" and should inherit from the UndergraduateCourse. The Registration class should have the following properties and methods:

- private Course course
- private int numStudents
- private int maxStudents
- private boolean isClosed
- void setCourse(Course course) - sets the course being registered
- void setNumStudents(int numStudents) - sets the number of students registered for the course
- void setMaxStudents(int maxStudents) - sets the maximum number of students that can register for the course
- void setIsClosed(boolean isClosed) - sets whether the course registration is closed or not
- void registerStudent() - registers a student for the course (if there is space available)

The program should create instances of the UndergraduateCourse and Registration classes, set their properties, and then register students for a course. Finally, the program should print out the details of the course, including the course code, course name, course credits, department/program name, and the number of students registered for the course. If the course is closed, print out a message indicating that the course is closed for registration.

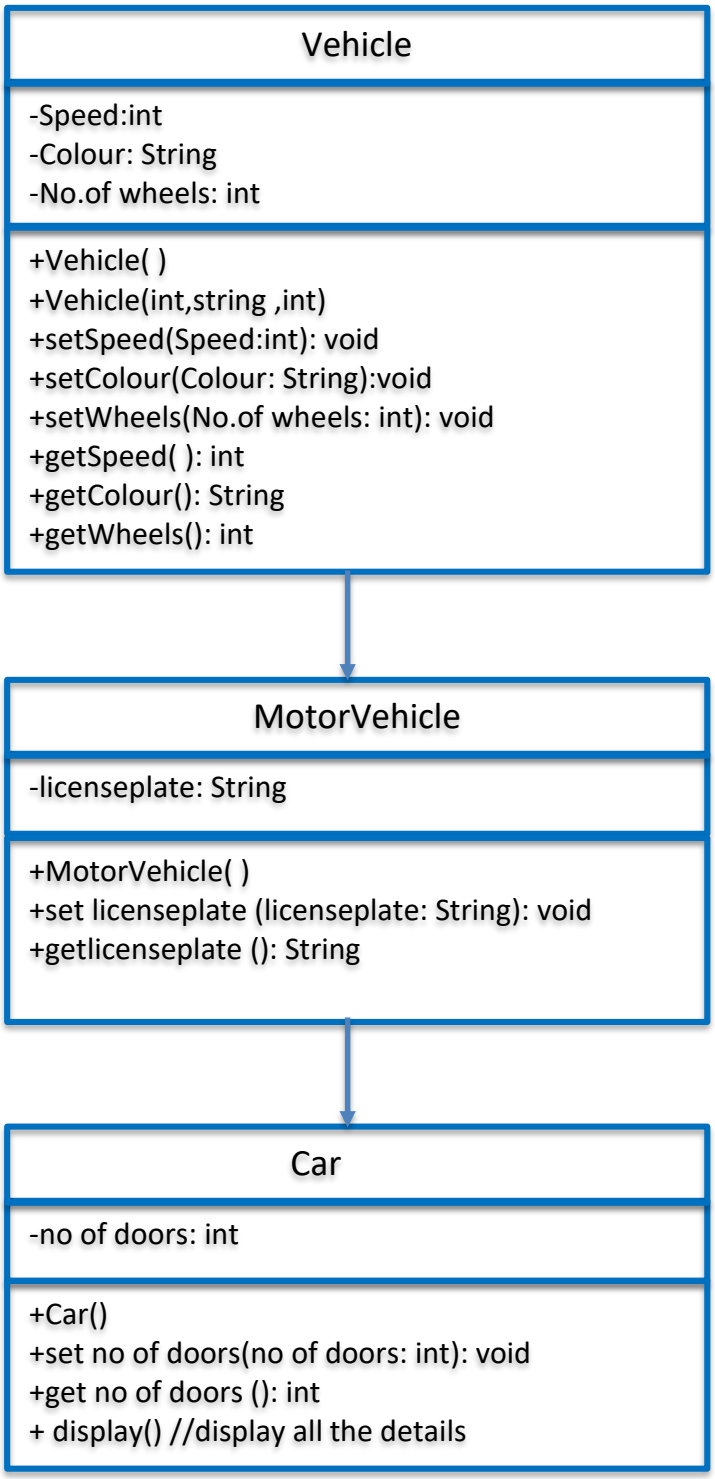
Task # 03:

Create a class named 'Rectangle' with two data members 'length' and 'breadth' and two methods to print the area and perimeter of the rectangle respectively. Its constructor having parameters for length and breadth is used to initialize length and breadth of the rectangle. Let class 'Square' inherit the 'Rectangle' class with its constructor having a parameter for its side (suppose s) calling the constructor of its parent class as super(). Print the area and perimeter of a rectangle and a square.

Task # 04:

A library wants to organize its system by categorizing books such as Java, C, C++, etc. Implement a program that contains a base class called Books that will contain members such as book ID, book name, book author, ISBN and price. All are protected members.

- Derive one class from the base class and name it as “Category1”.
- The class has one data member that is the category.
- Make a parameterized constructor and invoke the base class’s constructor.
- Create a display function and display all the details of the books in Category1 (show at least 3 books details).
- In the main program, perform object up casting and cast child object to a parent class object (Book).

Task # 05:

Implement the scenario given in the figure.